

Sparse-Autoencoder Dissection of Emotion Features in Frozen Speech Encoders

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Abstract

Linear probes show emotion is decodable in the middle-to-late layers of frozen speech transformers, but not which internal features carry it, nor whether the signal is acoustic or lexical. We train TopK Sparse Autoencoders (SAEs) on *per-frame* activations of six frozen large encoders and dissect their emotion features on the lexicon-controlled CREMA-D corpus (91 actors $\times$ 12 fixed sentences $\times$ 6 emotions). A confound-controlled selection—a feature’s firing must be more informative about emotion than about the spoken sentence or the speaker—isolates genuine emotion features from phonetic and speaker-driven ones. We find that (i) emotion *disentangles* into clean monosemantic features earlier (L2–L7) than where it is best linearly *decoded* (L10); (ii) all six encoders share an acoustic emotion-feature basis that generalizes across unseen speakers; and (iii) the recovered emotion signal is acoustic, not lexical. Decodability and disentanglement are thus distinct, and emotion in these encoders is carried by speaker-invariant acoustic features.

Keywords

speech emotion recognition, sparse autoencoders, mechanistic interpretability, probing, acoustic features

1 Introduction

Self-supervised speech encoders dominate speech emotion recognition, yet were trained on ASR-style objectives that encode linguistic structure. Probing reveals *where* emotion is readable, not *which* features encode it or whether the signal is acoustic or lexical. Sparse autoencoders (SAEs) decompose dense activations into sparse, near-monosemantic features [2, 5], moving the analysis from decodability to mechanism. The fixed lexicon of CREMA-D [3] dissociates acoustic delivery from text by construction, so any recovered emotion signal there is necessarily acoustic.

2 Method

We re-extract *un-pooled* per-frame hidden states from six frozen large encoders (HuBERT [6], WavLM, wav2vec2, w2v-BERT, MERT, Whisper [8]); the project’s cached embeddings are mean-pooled, too few to train an over-complete SAE. CREMA-D layer 13 yields 940,273 frames. We train a TopK SAE ($d_{in}=1024$, $d_{sae}=8192$, $k=32$, AuxK dead-feature revival), reaching FVU = 0.070 (93% variance explained), with 4/8192 dead features.

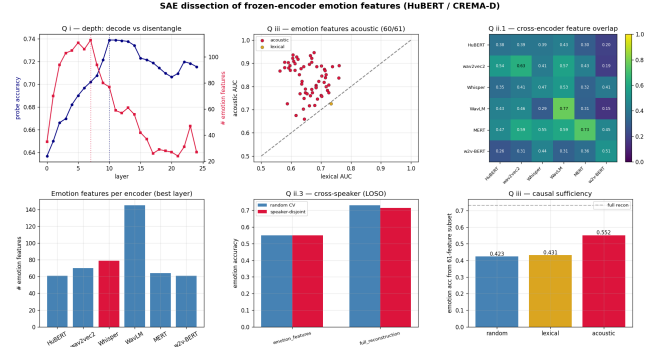


Figure 1: SAE dissection (HuBERT/CREMA-D). Top: depth decode-vs-disentangle; acoustic vs lexical; cross-encoder overlap. Bottom: emotion features per encoder; cross-speaker (leave-speaker-out); causal sufficiency (acoustic vs lexical vs random).

Emotion features are scored by the mutual information (MI) between a feature’s firing and the emotion label, thresholded at 5σ over a label-permutation null. **Confound control is decisive:** a feature qualifies only if $MI(\text{emotion})$ exceeds both $MI(\text{sentence})$ and $MI(\text{actor})$, and it is monosemantic (≥ 0.5 activation mass on one emotion). Significance and monosemanticity alone flag 6407 significant / 394 monosemantic features, but $\sim 66\%$ carry more sentence (phonetic) or speaker information than emotion—the documented “SAE rediscovers phonemes, not affect” failure [1]. After control, 61 genuine emotion features remain (median monosemanticity 0.59), dominated by anger and sadness; fear and neutral drop to zero. Acoustic-vs-lexical labeling uses cross-validated ROC-AUC of a logistic classifier predicting firing from eGeMAPS [4] (acoustic) vs. sentence identity (lexical); R^2 is unusable as the pooled features are $\sim 96\%$ sparse.

3 Results

3.1 Depth: decode $\neq$ disentangle (Q i)

A 25-layer sweep separates two curves (Fig. 1, top-left). Linear probe accuracy peaks at L10 (0.739); the count of monosemantic emotion features peaks earlier at L7, and monosemanticity at L2. Deep layers (L17–24) retain decodability (~ 0.72) but lose disentanglement (24–47 features): emotion stays linearly readable while becoming

Table 1: Key numbers (HuBERT / CREMA-D).

Quantity	Value
SAE FVU (variance explained)	0.070 (93%)
Dead features	4/8192
Emotion features (sig $\rightarrow$ mono $\rightarrow$ confound)	6407 $\rightarrow$ 394 $\rightarrow$ 61
Decode peak / disentangle peak layer	L10 / L7
Acoustic features (positive control)	60/61
Sufficiency acoustic / lexical / random	0.552/0.431/0.423
Cross-speaker leave-speaker-out drop	≈ 0

distributed. *Emotion organizes into clean features before the depth at which it is best decoded.*

3.2 Generalization across paradigms and speakers (Q ii)

All six encoders carry emotion features at their best layer (HuBERT 61, wav2vec2 70, Whisper 79, WavLM 145, MERT 64, w2v-BERT 61), all anger/sadness-dominated. Matching features across encoders by per-utterance firing correlation, Whisper $\rightarrow$ SSL overlap (0.40) $\approx$ SSL $\rightarrow$ SSL overlap (0.39): on fixed-lexicon CREMA-D Whisper is *not* distinct, its known divergence concerning the lexical shortcut rather than acoustic emotion encoding. Under speaker-disjoint cross-validation over 91 actors, emotion accuracy is unchanged (random 0.551 vs. disjoint 0.551; leave-speaker-out drop ≈ 0) and each feature fires for $\sim 95\%$ of its emotion’s actors. Because selection excluded speaker-entangled features, the published MESD leave-one-speaker-out collapse (~ 0.30) is attributable to exactly those excluded features.

3.3 Acoustic, not lexical (Q iii)

The positive control (fixed lexicon $\Rightarrow$ must be acoustic) labels 60/61 emotion features acoustic (eGeMAPS firing-AUC 0.83 vs. sentence AUC 0.66). Causally (Fig. 1, bottom-right), emotion decodes from the 61 acoustic features at 0.552, vs. 0.431 from lexical and 0.423 from random features (6-class; full reconstruction 0.73). The recovered emotion signal is acoustic.

4 Discussion and Limitations

Decodability and disentanglement are distinct axes: a probe reads emotion best where the representation is linearly separable, but the encoder packs emotion into clean monosemantic features earlier. On lexicon-controlled data all encoders converge on a shared acoustic basis whose features survive speaker-disjoint evaluation; the Whisper-vs-SSL divergence is about exploiting a lexical shortcut when one exists [7], which CREMA-D cannot pose and which the deferred EMIS experiment targets. Utterance-pooled sparse features make necessity-ablation of small subsets near-null (the signal is distributed), so sufficiency is the informative test. The IEMOCAP real-transcript ablation, the SpeechCraft cross-lingual arm, and the EMIS audio-vs-text separation are deferred (data unavailable on our cluster) with runnable stubs.

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